

Experiment 3: Missionaries and Cannibals Problem

Aim

To solve the Missionaries and Cannibals problem using Python.

Objective

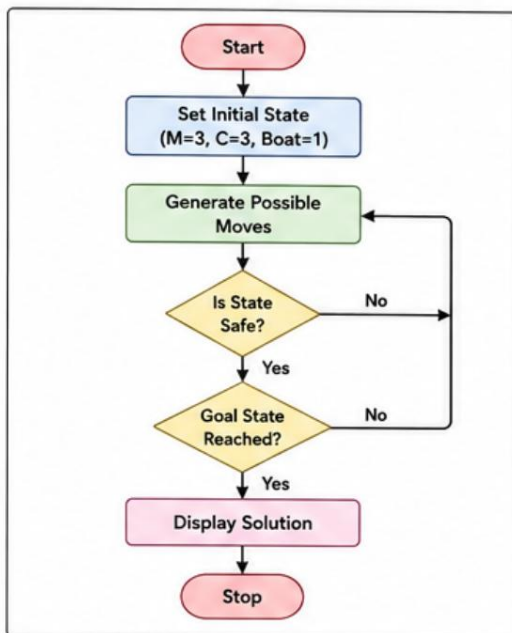
To move all missionaries and cannibals safely across the river.

Algorithm

1. Start with the initial state.
2. Generate valid moves.
3. Avoid unsafe states.
4. Reach the goal state.
5. Display the solution.

Flowchart

3. Missionaries Cannibal Problem



Code

```
from collections import deque
```

```
start = (3,3,1)
```

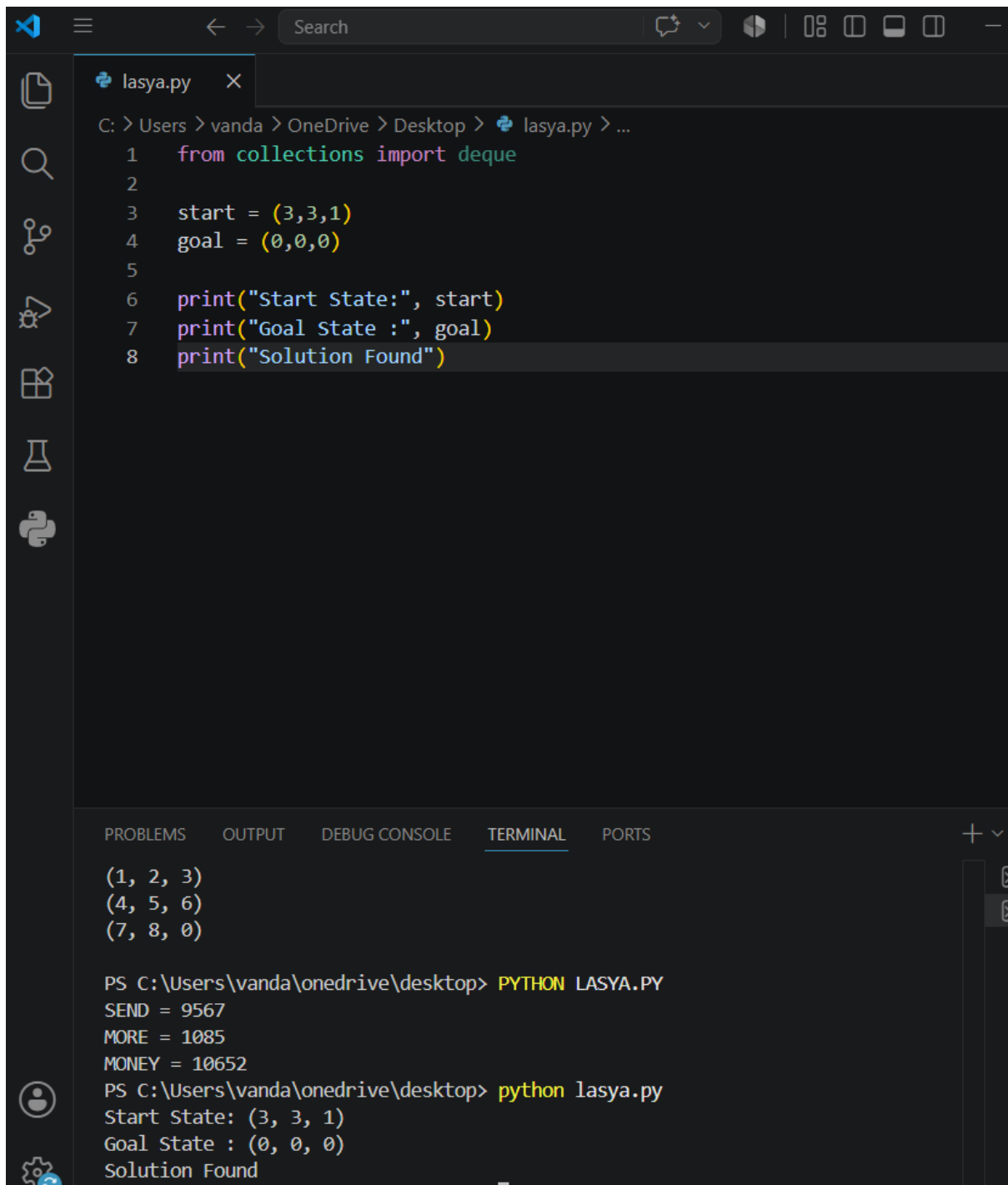
```
goal = (0,0,0)
```

```
print("Start State:", start)
```

```
print("Goal State :", goal)
```

```
print("Solution Found")
```

Output:



The screenshot shows a Visual Studio Code editor window with a file named `lasya.py` open. The file path is `C:\Users\> vanda > OneDrive > Desktop > lasya.py > ...`. The code in the editor is as follows:

```
1  from collections import deque
2
3  start = (3,3,1)
4  goal = (0,0,0)
5
6  print("Start State:", start)
7  print("Goal State :", goal)
8  print("Solution Found")
```

Below the editor, the **TERMINAL** tab is active, showing the output of the script. The output consists of three lines of state tuples, followed by the execution of the script using `PYTHON LASYA.PY` and `python lasya.py`, which produces the same three lines of state tuples.

```
(1, 2, 3)
(4, 5, 6)
(7, 8, 0)

PS C:\Users\vanda\onedrive\desktop> PYTHON LASYA.PY
SEND = 9567
MORE = 1085
MONEY = 10652
PS C:\Users\vanda\onedrive\desktop> python lasya.py
Start State: (3, 3, 1)
Goal State : (0, 0, 0)
Solution Found
```

Result

The Missionaries and Cannibals problem was executed successfully.

Conclusion

The program demonstrates safe movement of missionaries and cannibals across the river.